16 June 1970
SUPERSEDING
MIL-T-11455A
15 March 1953

MILITARY SPECIFICATION

TEST SET, BATTERY AN/USM-63( )/U

1. SCOPE

1.1 This specification covers one type of battery tester designated as Test Set, Battery, AN/USM-63( ) consisting of Test Set, Battery TS-737( )/U, and Resistor Assembly MX-1678( )/U.

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect at date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communications Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-T-55549	Test Set, Battery AN/USM-63 Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-810	Environmental Test Methods

FSC 6625

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DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester
SC-DL-76244	Test Set, Battery TS-737( )/U
SC-DL-104537	Resistor Assembly MX-1678( )/U

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description. Test Set, Battery AN/USM-63( ) consists of Test Set, Battery TS-737( )/U and Resistor Assembly MX-1678( )/U.

3.1.1 Test Set, Battery TS-737( )/U. Test Set, Battery TS-737( )/U is a portable, self contained, high precision battery tester. It consists of a dual range voltmeter, hermetically sealed, with the meter face calibrated in two voltage scales for measuring cell voltage and battery voltage. The test set is equipped with a fixed load resistor and toggle switch, to permit measurements of batteries under load and no load conditions. Cell voltages are measured by means of a retractable rubber cord to which is affixed a special test prod.

3.1.2 Resistor Assembly MX-1678( )/U. Resistor Assembly MX-1678( )/U is a portable, self-contained unit. A ventilated metal case houses two load resistors. A pair of test leads are stored in the control box which are used for test purposes and for discharging nickel-cadmium cells and batteries.

3.2 First article samples. The contractor shall furnish two (2) complete first article samples for approval provided the invitation for bid and the contract requires first article samples.

3.3 Construction. The equipment shall be constructed in accordance with the following drawings and the requirements specified herein:

SC-DL-76244	-	Test Set, Battery TS-737( )/U
SC-DL-104537	-	Resistor Assembly MX-1678( )/U

3.4 Parts, materials and processes - general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection of requirements therein, shall apply.

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3.5 Finish, protective. The equipment shall be finished in accordance with MIL-F-14072. This includes finish of hardware; such as screws, etc., and necessary touch-up after mounting.

3.6 Marking. The marking shall conform to MIL-M-13231.

3.6.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings shall be readily visible with a minimum disassembly of the equipment.

3.6.2 Serial numbers. Serial numbers are required on each nomenclatured and type designated item, except cords and cables.

3.7 Electrical.

3.7.1 Accuracy. The Test Set, Battery TS-737( )/U voltmeter at an ambient temperature of 77° ± 5°F. shall be accurate to ± 0.5 percent of indicated value within the ranges of 1.0 to 1.4 volts and 25.2 to 30.0 volts. All other points shall be accurate to ± 1.0 percent of full scale value, when tested in accordance with 4.7.1.

3.7.2 Resistance. The Test Set, Battery TS-737( )/U resistance value across test set terminals "E 2" and "E 3" shall be 2.8 ohms ± 5 percent when tested in accordance with 4.7.2.

3.7.3 Operational. The Test Set, Battery TS-737( )/U shall meet the requirements of 4.7.3.(f) when tested in accordance with 4.7.3.

3.7.4 Plus cell. The Resistor Assembly MX-1678( )/U resistance from negative binding post to binding post marked "PLUS CELL" shall be 0.117 ohms ± 5 percent when the switch is in cell load position when tested in accordance with 4.7.4.

3.7.5 Plus battery. The Resistor Assembly MX-1678( )/U resistance from the negative binding post to binding post marked "PLUS BATTERY" shall be 2.8 ohms ± 5 percent when the switch is in load position, when tested in accordance with 4.7.5.

3.7.6 Accuracy at temperature. The Test Set, Battery TS-737( )/U accuracy of the voltmeter at -4°F. and 125°F. shall be ± 1.0 percent of indicated value within the ranges of 1.0 to 1.4 volts and 25.2 to 30.0 volts, all other points shall be ± 2 percent of full scale when tested in accordance with 4.7.1.

3.8 Service conditions. The Test Set, Battery AN/USM-63( ) shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered, no movable part of control shall be shifted in setting, position or adjustment, and no permanent degradation shall be caused in performance below that specified in the following service conditions.

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3.8.1 Low temperature. The test set shall be subjected to the low temperature extremes of -40°F. operating and -65°F. storage and shall meet the electrical requirements of 3.7.1, 3.7.2, 3.7.3, 3.7.4, 3.7.5 and 3.7.6 at times specified when tested in accordance with 4.8.1.

3.8.2 High temperature. The test set shall be subjected to the high temperature extremes of 125°F. operating and 160°F. storage and shall meet the electrical requirements of 3.7.1, 3.7.2, 3.7.3, 3.7.4, 3.7.5 and 3.7.6 at times specified when tested in accordance with 4.8.2.

3.8.3 Humidity. The test set shall exhibit no physical damage of parts and materiel, such as corrosion, rust, blistering, swelling, deterioration, etc.; or electrical degradation, and shall meet the requirements of 3.7.1, 3.7.2, 3.7.3, 3.7.4 and 3.7.5 at times specified, when tested in accordance with 4.8.3.

3.8.4 Fungus. The test set shall show no evidence of viable fungus when tested in accordance with 4.8.4.

3.9 Visual and mechanical. The Test Set, Battery AN/USM-63( ) shall meet the requirements of MIL-STD-252 as specified in 4.9.

3.10 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or the equipment. Individual items shall not be hand picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc. (see 4.10).

3.11 Bounce preconditioning. Each test set produced shall be subjected to the bounce preconditioning test of 4.6 prior to Group A inspection. After bounce preconditioning, the test set shall not be repaired, aligned, cleaned or otherwise changed prior to Group A acceptance test.

3.12 Workmanship. The test set shall be manufactured and assembled in accordance with the applicable portions of MIL-P-11268 as follows:

General requirements for plastic material and parts.
Wiring and cabling including:
- Slack.
- Clearance.
- Splicing.
- Connectors, general.

Flux and cleaning agents for soldering.
Process for soldering.
Cleaning of equipment.
General process for securing of parts.
Process of welding.
Connectors, general.
Mechanical operation of controls.
Tropicalization of material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the contract and specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (does not include preparation for delivery) (see 4.3).

(b) Inspection covered by subsidiary documents (see 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.11).

4.3 First article inspection. This inspection shall be performed by the contractor and witnessed by a Government representative unless otherwise specified in the contract. It shall consist of Group A, B and C inspections (see Tables I, II and III, respectively) on the two (2) samples of the completely assembled Test Set, Battery AN/USM-63( ). The first article samples shall be inspected in this order: Group A and B for all units; and designated units for specific Group C inspections. Designation of units to specific Group C inspections shall be determined by the Government production engineer. After completion of all Group C tests, conforming units must be reinspected and pass all Group A inspections.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.5
Marking	3.6
Workmanship	3.12

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4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit produced shall be inspected by the contractor for conformance to all the Group A requirements and tests in Table I. Each lot shall be subject to Government verification and this inspection, including sampling, shall conform to Table I and the ordinary inspection procedures of MIL-STD-105.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table I - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.9	4.9	1.5dphu*	4.0dphu
Electrical				
Accuracy	3.7.1	4.7.1	1.5dphu for the group	**
Resistance	3.7.2	4.7.2		
Operation	3.7.3	4.7.3		
Plus cell	3.7.4	4.7.4		
Plus battery	3.7.5	4.7.5		

* Defects per hundred units.
** All electrical tests are major.

4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table II and to the special procedures for small sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed Group A inspection, and on samples selected from units that have been subjected to and met the Group A inspection.

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4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group B inspection

Inspection	Req. Para.	Test Para.	AQL
Interchangeability	3.10	4.10	6.5dphu

* Defects per hundred units.

4.5.3 Group C inspection. This inspection shall consist of the inspection specified in Table III and shall be performed on sample units that have been subjected to and met Group A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for inspection.

4.5.3.1.1 Two (2) samples of complete equipment shall be selected at random, one (1) sample for each Group 1 and Group 2 in Table III. The samples selected shall be at the start of the contract from the first quality conformance inspection lot. These samples shall constitute Group C requirements for first 200 units produced.

4.5.3.1.2 Thereafter, one (1) sample of complete equipment shall be selected at random for Group 1 in Table III from every 200 units and fraction thereof produced.

4.5.3.2 Evaluation of samples submitted to individual Group C inspection. After referenced samples in 4.5.3.1 have been subjected to an individual Group C inspection test and prior to subjection to the next Group C test, the sample shall meet the requirements of 3.7.1, 3.7.2, 3.7.3, 3.7.4 and 3.7.5.

4.5.3.3 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be

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subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government). The final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.3.4 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

Table III - Group C inspection

Inspection	Req. Para.	Test Para.
Group 1		
Low temperature	3.8.1	4.8.1
High temperature	3.8.2	4.8.2
Humidity	3.8.3	4.8.3
Group 2		
Fungus	3.8.4	4.8.4

4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

4.5.5 Disposition of Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order provided units meet requirements of 4.5.4.

4.6 Bounce preconditioning. Each test set produced, shall be placed in its normal operating position on the table of a package tester of suitable capacity as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 1000SC and SC-A-46439, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment under test to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute and shall meet the requirements of 3.11.

4.7 Electrical.

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4.7.1 Accuracy. The Test Set, Battery TS-737( )/U voltmeter shall be tested at the 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 25.2, 28.0, 30.0, 32.0, 33.6, and 34.1 voltage indications by comparison checks with a standard voltmeter which has an accuracy of at least ± 0.05 percent. The test set voltmeter shall meet the requirements of 3.7.1 or 3.7.6 as applicable.

4.7.2 Resistance.

(a) The Test Set, Battery TS-737( )/U switch shall be closed-circuited.

(b) Using a standard resistance bridge with an accuracy of at least 0.05 percent, measure the resistance across test set terminals "E 2" and "E 3". It shall meet the requirements of 3.7.2.

4.7.3 Operational.

(a) The Test Set, Battery TS-737( )/U switch shall be in the open-circuit position.

(b) Apply a constant voltage source of 28 volts capable of delivering 10 amperes to test set terminals "E 2" and "E 3".

(c) Read the value indicated on the test set voltmeter.

(d) The test set switch shall then be operated to the closed-circuit position for a period of one (1) minute.

(e) After one (1) minute has elapsed the test set switch shall be operated to the open-circuit position.

(f) The value indicated on the test set voltmeter shall not change from the initial value read on the test set in 4.7.3.(c) (see 3.7.3).

4.7.4 Plus cell. The Resistor Assembly MX-1678( )/U resistance from negative binding post to binding post marked "PLUS CELL" shall be measured using a standard resistance bridge with an accuracy of 0.05 percent, and shall meet the requirements of 3.7.4.

4.7.5 Plus battery. The Resistor Assembly MX-1678( )/U resistance from the negative binding post to binding post marked "PLUS BATTERY" shall be measured using a standard resistance bridge with an accuracy of 0.05 percent, and shall meet the requirements of 3.7.5.

4.8 Service conditions. The Test Set, Battery AN/USM-63( ) shall be tested in accordance with the following service conditions.

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4.8.1 Low temperature. The test set in its normal operating configuration shall be subjected to the low temperature test specified in Procedure I, Method 502 of MIL-STD-810, with temperatures in Step 2 held for 17 hours and in Step 4 held for 4 hours, and shall meet the requirements of 3.8.1.

4.8.2 High temperature. The test set in its normal operating configuration shall be subjected to the high temperature test specified in Procedure I, Method 501 of MIL-STD-810, with temperature in Step 4 held for 4 hours, and the temperature in Step 2 held for 16 hours, in lieu of the 48 hours stated in MIL-STD-810. The test set shall meet the requirement of 3.8.2.

4.8.3 Humidity. The test set in its normal operating configuration shall be subjected to the humidity test specified in Procedure II, Method 507 of MIL-STD-810, and shall meet the requirements of 3.8.3.

4.8.4 Fungus. The test set in its normal operating configuration shall be subjected to 28 days of the fungus test specified in Procedure I, Method 508 of MIL-STD-810 and shall meet the requirements of 3.8.4.

4.9 Visual and mechanical. The test set shall be examined for the defects listed in MIL-STD-252 and shall meet the requirements of 3.9.

4.10 Interchangeability. The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers and mountings.

(d) Location of battery holder and terminals.

(e) Location of switches, sockets and fasteners.

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(f) Location of cover and fasteners on carrying case which must mate.

(g) Location of panel mounting screw holes on basic unit for mating with case.

4.11 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the requirements stated in MIL-T-55549.

6. NOTES

6.1 Intended use. The Test Set, Battery AN/USM-63( ) is used to test the terminal and cell voltage of nickel-cadmium storage batteries under load, under no load and during charge conditions. In addition, the battery test set is used to discharge a nickel-cadmium cell or a storage battery.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (Levels A, B or C) (see section 5).

(c) If rough handling and functional tests are required.

(d) Quantity of first articles required.

(e) Place of final inspection.

(f) Technical literature required.

(g) Quantity of tools and running spare parts required.

(h) Marking for shipment and shipping containers.

6.3 Nomenclature and type designation. The contractor should apply for nomenclature and type designation in accordance with the applicable clause in the contract. The parentheses in the type designation will be deleted or replaced by a letter identifying the particular design for example: SG-12F/U or AN/URM-73C.

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6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3)

6.5 Temperature conversion. Temperature conversion from fahrenheit to centigrade as listed in this specification are as follows:

160°F	=	71°C
125°F	=	51.5°C
-4°F	=	-20°C
-65°F	=	-53.9°C

6.6 Verification inspection. Verification inspection by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIANS:

ARMY-EL

User Activities:
ARMY-EL, ME, MU
NAVY - AS, MC, OS, SH

Preparing Activity

ARMY-EL

Project Nr. 6625-0486

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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